

Benjamin Schmid Ties

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Education

B.Sc.	Free University of Bolzano , Electronics and Cyber-Physical Systems Engineering	Bolzano, Italy
	• Final grade: 108/110 • Thesis: Wearable Screen-Printed DC System for Iontophoretic Sweat Extraction in Biomedical Applications  (Short form)	09/2022 - 07/2025
Exchange	Technical University of Munich , Electrical Engineering and Information Technology	Munich, Germany
	• GPA: 1.7 • Courses: Deep Learning, Electronic Circuits, Computer Networks	04/2024 - 09/2024
Maturità	Technical High School „Max Valier“ , Electronics and Automation	Bolzano, Italy
	• Final grade: 84/100	09/2017 - 07/2022

Experience

ewo , Working Student/Freelancer	Cortaccia, Italy
Developed an IoT device for object detection in street lighting	09/2024 - present
• Designed, trained and deployed a neural network for object detection on an ESP32-S3 microcontroller with highly constrained resources • Built a custom Linux distribution for a Raspberry Pi using Yocto, including support for wireless firmware updates and a web server displaying real-time object detections	
Alpitronic , Working Student	Bolzano, Italy
Worked on hymaint, a maintenance platform for electric vehicle chargers	10/2023 - 03/2024
• Developed Python scripts for troubleshooting and configuring Hypercharger • Optimized central library component, decreasing CAN-Bus communication latency by 92%	
KERR , Mandatory Internship	Bolzano, Italy
• Developed a C library for BLE communication on ESP32-S3 microcontroller • Configured a GPRS/LTE connection using a u-blox modem in an IoT device • Replaced soldered components on prototype boards	07/2023 - 08/2023
Alpitronic , Internship	Bolzano, Italy
• Developed a Proxy Server in C++ with ZMQ and WebSocket interface • Created a command-line tool in C++ to generate OCPP messages interactively • Assisted in setting up a GitLab CI/CD pipeline	07/2022 - 09/2022
Progress Maschinen & Automation , Internship	Bressanone, Italy
• Designed user interfaces for industrial machines using C# WinForms • Automated the generation of error reports to accelerate customer support and reduce downtime • Diagnosed and fixed bugs in proprietary CAD software	07/2021 - 08/2021

Projects

echessboard (Electronic chessboard) 
launchpad (Portable handheld drum machine) 
Cartesian Robot with computer vision for automated object sorting (High School final project) 
Implementation of Frechet Distance Algorithm in C including interactive application 
Movie2Ascii (Convert a video to ASCII art in a multi-threaded C++ application) 

Skills

Programming: C, Python, Git, VHDL, C++, Bash, C#, CMake, Rust, MATLAB
Technologies: Microcontrollers, FPGA, PCB design, Linux, Soldering, Yocto, Docker, 3D printing, Machine Learning, LaTeX
Languages: German (native), English (fluent), Italian (fluent)

Other

• Participated in BR41N.IO hackathon in 2023, where our team developed a drone controlled by a Brain-Computer Interface (BCI) • Participated in the Competitive Programming Team at the Free University of Bolzano • Contributions to open-source projects, including raygui
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